



Realistic preference & constraint model

1. Core User Preference Dimensions

primary drivers of itinerary quality

1.1 Travel Style

- Relaxed vs Fast-paced
- Explorer vs Comfort-seeker
- Structured vs Spontaneous
- Popular attractions vs Hidden gems
- Guided vs Self-guided

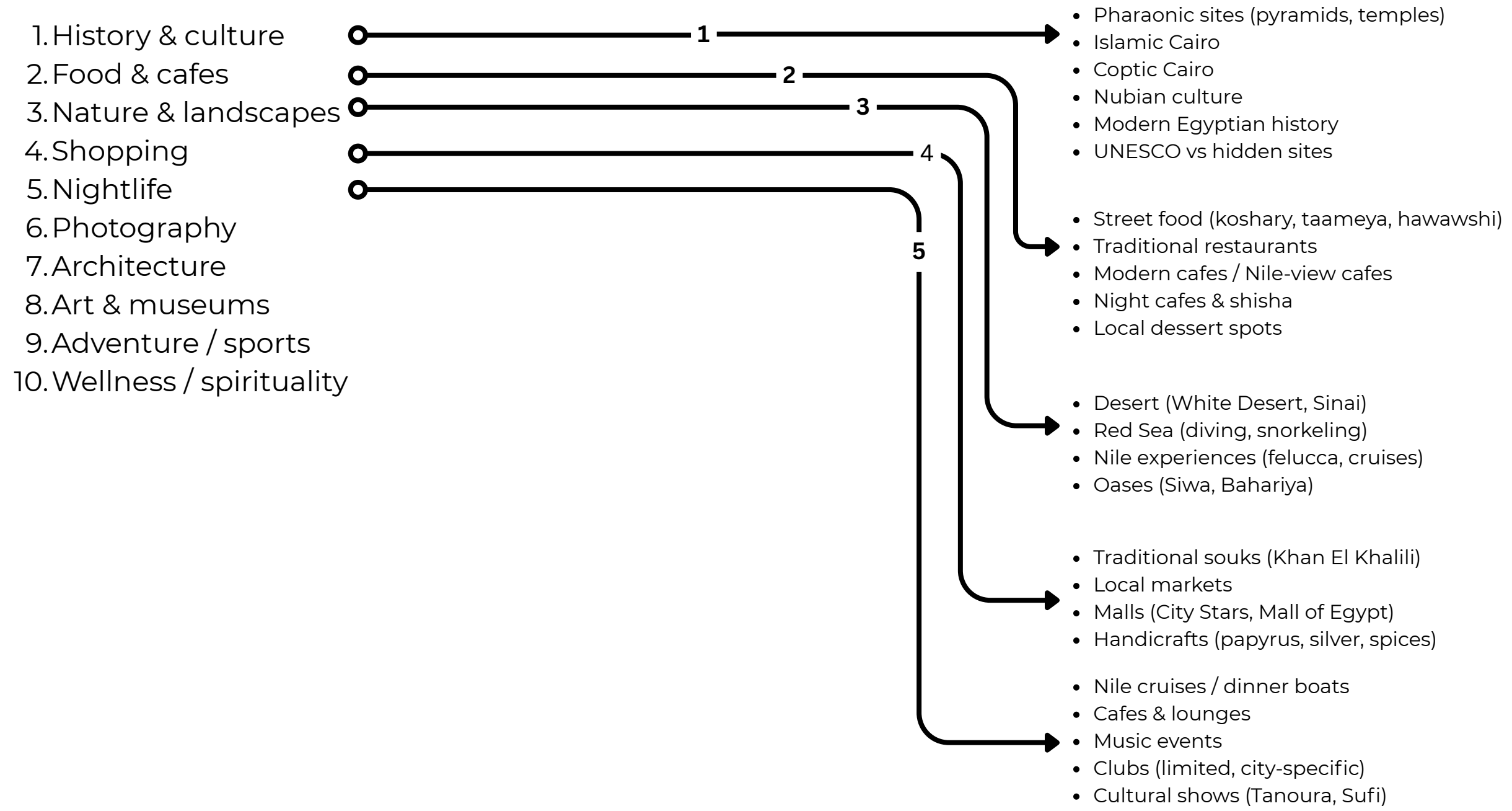
1.2 Interests & Motivations (Weighted)

Each interest should be a 0–1 or 0–5 score

```
"interests": {  
  "food": 0.9,  
  "history": 0.7,  
  "nature": 0.4 }  

```

1.2.1 Primary Interest Categories (anchors)



1.3 Budget Preferences (Soft + Hard)

- Total trip budget
- Daily spend comfort level
- Willingness to splurge (yes/no)
- Category sensitivity:
 - Food
 - Transport
 - Activities
 - Accommodation



```
"budget": {  
  "daily_limit": 120,  
  "food_priority": "high",  
  "luxury_events": false}
```

1.4 Time & Energy Constraints

- Start/end time per day
- Max activities per day
- Walking tolerance (km or minutes)
- Preferred break frequency
- Jet lag / fatigue sensitivity



```
"energy_profile": {  
  "max_daily_hours": 8,  
  "walking_limit_km": 6,  
  "break_every_hours": 2}
```

2. Travel Logistics & Constraints

2.1 Dates & Trip Structure

- Arrival & departure times
- Number of days
- Fixed bookings (hotels, events)
- Day trips vs multi-city

2.2 Location Constraints

- Starting point each day
- Accommodation location
- Must-visit places
- Must-avoid places

2.3 Transport Preferences

- Walking / Public transport / Taxi / Car
- Max travel time between activities
- Cost sensitivity
- Accessibility needs

3. Social & Group Dynamics

3.1 Group Type

- Solo
- Couple
- Friends
- Family
- Kids / Elderly present

3.2 Personality Mix (Advanced)

- Planner vs Explorer
- Early bird vs Night owl
- Risk-averse vs Adventure-seeking

4. Contextual & Environmental Preferences

4.1 Crowd Tolerance

- Avoid peak hours
- Prefer quiet locations
- Comfortable with tourist density

4.2 Climate & Comfort

- Heat tolerance
- Indoor vs outdoor preference
- Seasonal awareness

4.3 Safety & Cultural Sensitivity

- Neighborhood safety preference
- Cultural norms comfort level
- Language barrier tolerance

5. Content & Experience Quality Preferences

5.1 Experience Depth

- Quick highlights vs Deep dive
- Photo stops vs Educational tours
- Passive vs Interactive

5.2 Authenticity Preference

- Local experiences
- Tourist-friendly
- Mixed

6. AI-Specific Preference Modeling

6.1 Feedback Loops

The algorithm must learn from:

- Skipped activities
- Time overruns
- User edits
- Ratings after each day

6.2 Flexibility Level

How much the plan is allowed to change.

- Fixed plan
- Suggest alternatives
- Real-time replanning

6.3 Explanation Preference

- Just itinerary
- With reasoning
- With local tips

7. Algorithmic Decision Factors (Behind the Scenes)

Your AI should optimize across:

- Preference match score
- Travel time minimization
- Energy balance
- Budget adherence
- Opening hours feasibility
- Weather awareness
- Crowd prediction
- User satisfaction probability

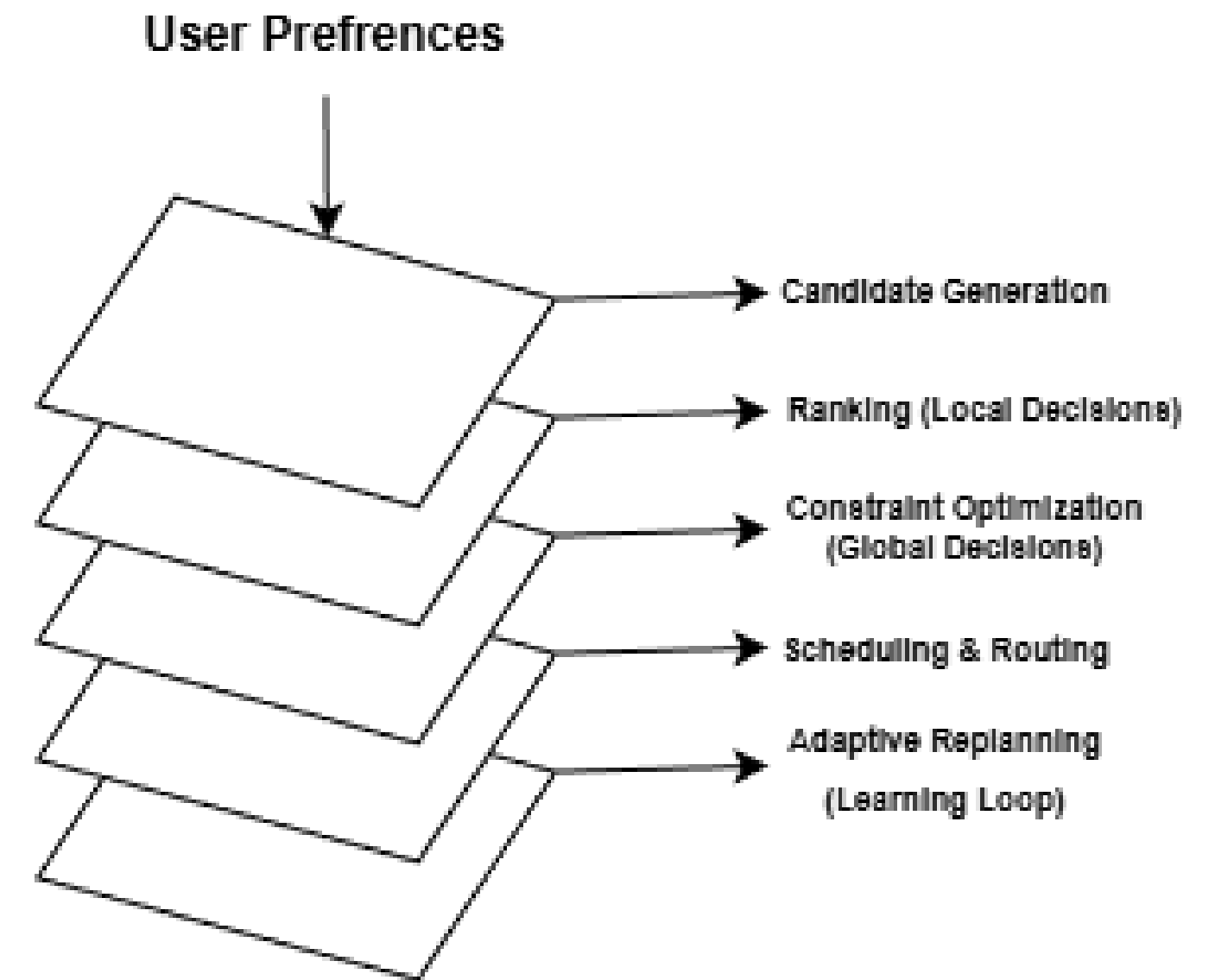
8. Minimum Viable Preference Set (MVP)

If you want a strong MVP, collect:

1. Interests (weighted)
2. Budget comfort
3. Travel style
4. Daily energy limit
5. Must-see locations
6. Transport preference
7. Group type

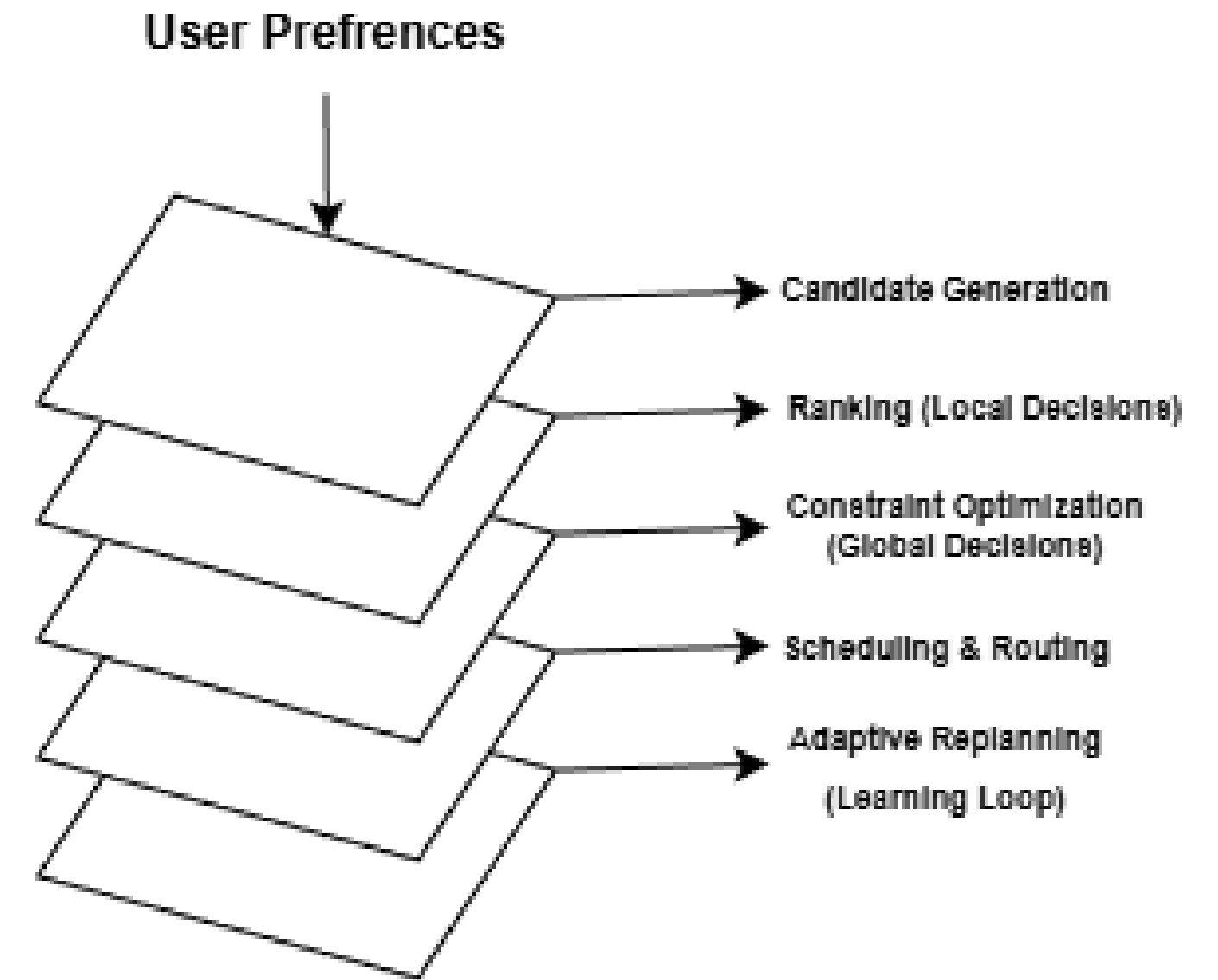
Candidate Generation (NLP + Retrieval)

- Semantic search (embeddings)
 - Geo-based filtering
 - Category inference
 - Review text analysis
-
- LLMs (to classify & enrich places)
 - Embedding models (similarity search)



Ranking Models (Local Optimization)

- Learning-to-Rank models
- Gradient boosting
- Neural ranking

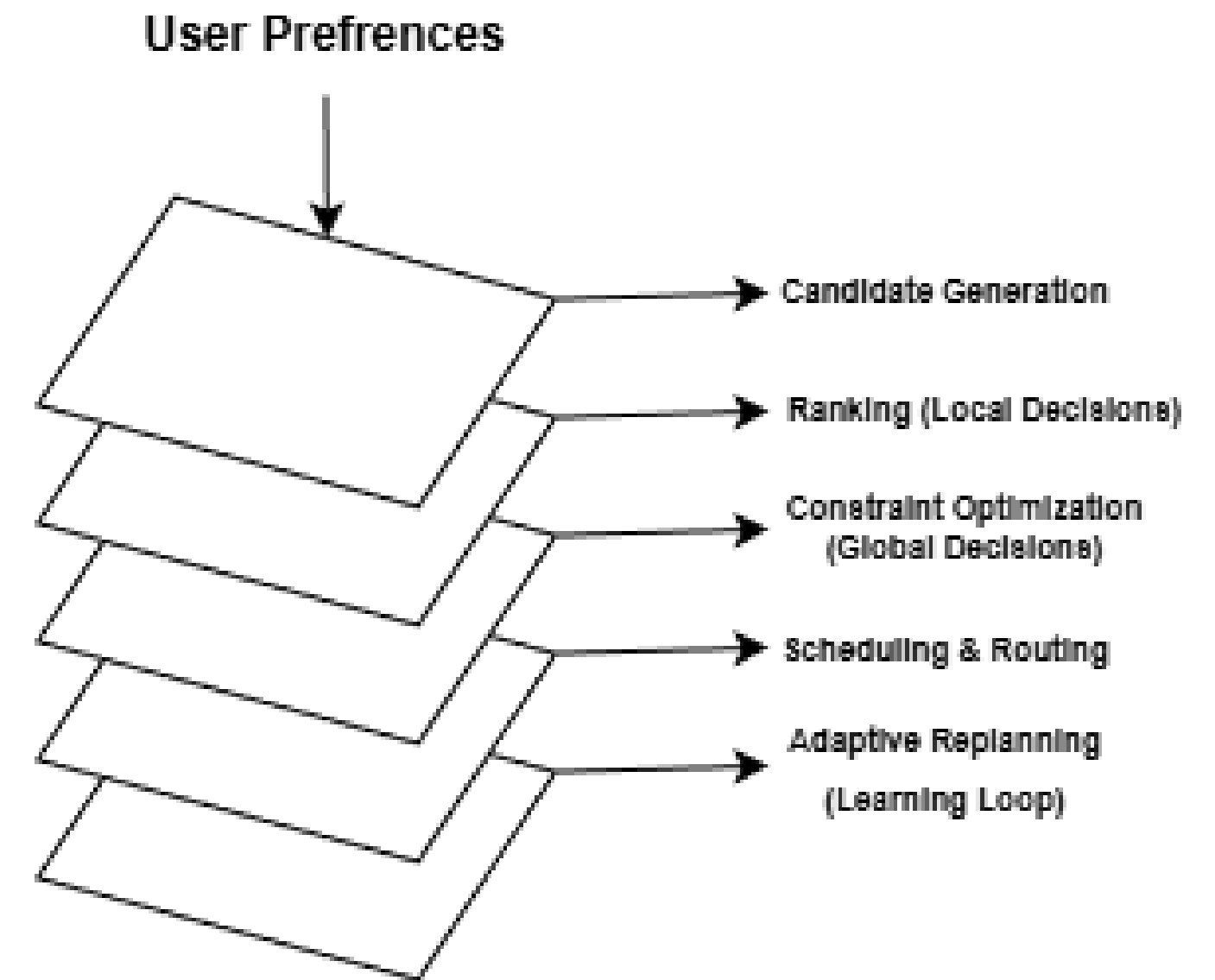


Constraint Satisfaction & Optimization (This Is the Core)

- Constraint Satisfaction Problems (CSP)
- Integer Linear Programming (ILP)
- Heuristic search
- Genetic Algorithms
- Simulated annealing

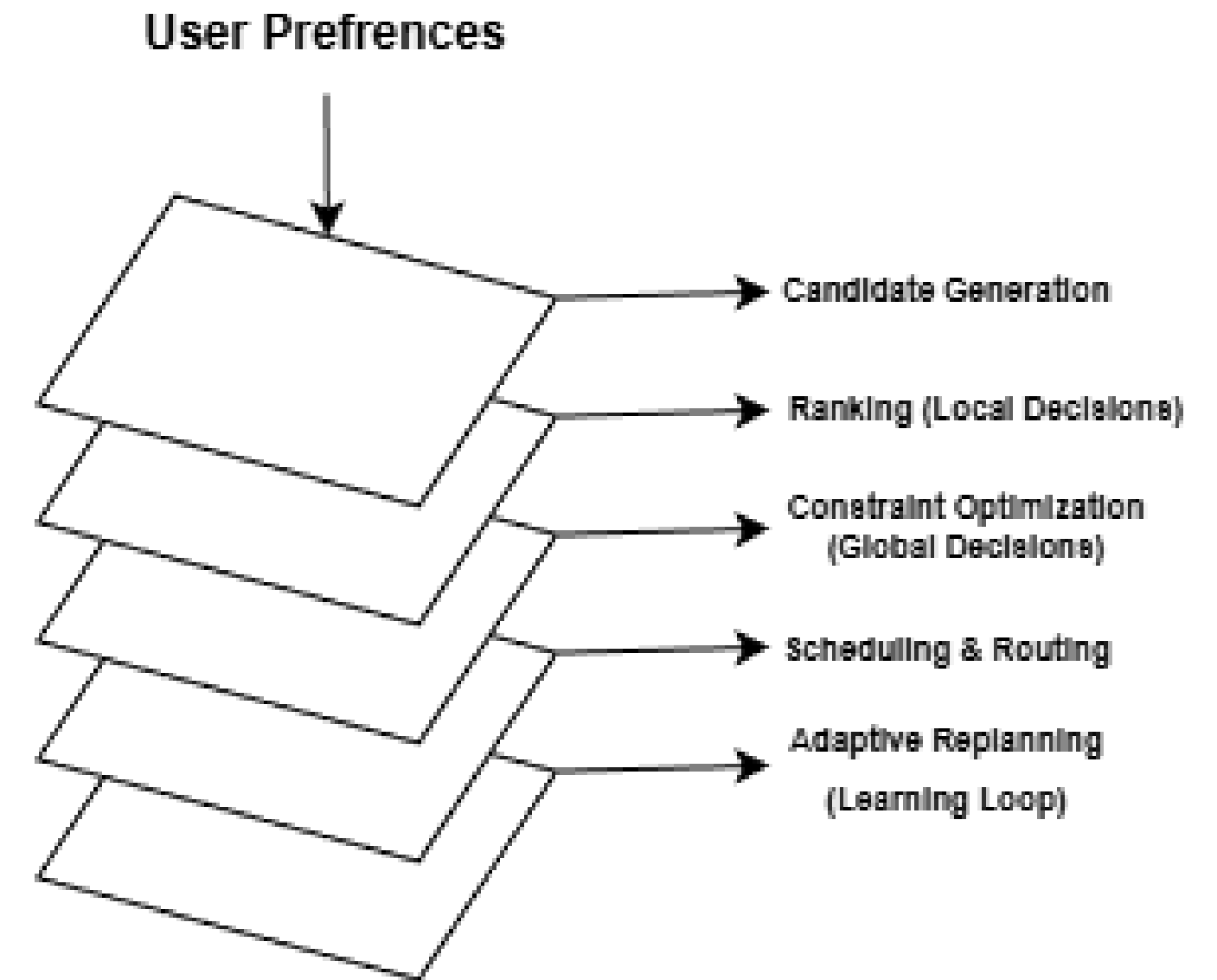
Constraints:

- Opening hours
- Travel time
- Energy limits
- Budget caps
- Fixed bookings



Routing & Scheduling (Graph Algorithms)

- Shortest path (Dijkstra / A*)
 - Time-dependent routing
 - Traveling Salesman Problem (TSP variants)
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- Minimal backtracking
 - Logical geographic flow
 - Real travel feasibility



Memories summary

- Memory Timeline
- Day by Day Images to support that

Events

- Tazkerti
- scenenow

Maps

- CARTO → analyze attractions, density, travel time
- AI / LLM → optimize schedule & preferences
- Mapbox → interactive mobile map

Points of Interest (POIs)

Fields needed:

- Name
- Latitude / Longitude
- Category
- Sub-category
- Estimated visit duration
- Entry cost (EGP)
- Opening hours
- Indoor / outdoor
- Family-friendly (yes/no)